```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load the Iris dataset
df = pd.read_csv("./datasets/iris.csv")

# Select features (attributes) for clustering (e.g., sepal_length,
        sepal_width, petal_length, petal_width)
X = df.iloc[:, 1:-1]  # Exclude the first column (id) and the last column
        (species)

# Standardize the feature matrix (important for K-Means)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Determine the optimal number of clusters using the elbow method
inertia = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)
```

```
D:\Python\lib\site-packages\sklearn\cluster\_kmeans.py:870: FutureWarning:
The default value of `n_init` will change from 10 to 'auto' in 1.4. Set the
value of `n_init` explicitly to suppress the warning
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```
print(inertia)
```

```
[600.0, 223.73200573676345, 140.96581663074699, 114.38954350297054,
91.00244098567246, 80.24972235577131, 71.8094621471561, 62.641008351188034,
54.25747690371483, 49.958828372190034]
```

```
# Plot the elbow curve
plt.figure(figsize=(8, 6))
plt.plot(range(1, 11), inertia, marker='o', linestyle='--')
plt.xlabel('Number of Clusters')
plt.ylabel('Inertia (Within-cluster Sum of Squares)')
plt.title('Elbow Method for Optimal Number of Clusters')
plt.grid()
plt.show()
```

